

3. The tables show the molecular formulae of some alkanes and alkenes.

Alkanes
CH_4
C_2H_6
C_3H_8
C_4H_{10}

Alkenes
C_2H_4
C_3H_6

- (a) The general formula for the alkene family is C_nH_{2n} . Give the general formula for the alkane family. [1]

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- (b) When alkanes and alkenes completely burn in air they form the same two products.

Give the chemical formulae for both products. [1]

..... and

- (c) Draw the structural formula for propene. [1]

- (d) Bromine water is used to distinguish alkenes from alkanes. Describe the colour **change** seen when bromine water is added to an alkene. [1]

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